



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	FEBRUARY 2024
GRADE	11	MARKS	40
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	40 MINUTES
CLASS			

TEST 2

INSTRUCTIONS:

1. This paper consists of 7 pages- 6 pages of questions and 1 page of additional writing space. Please check that your question paper is complete.
2. Answer all questions **on the question paper**.
3. Write in blue or black pen only.
4. Read all questions carefully and answer in as much detail as possible.
5. All diagrams should have a clear heading.
6. It is in your own interest to write legibly and to present your work neatly.

QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	14	
3	16	
TOTAL: 40		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question in the space provided in the table below: (5)

1.1.1.	1.1.2.	1.1.3.	1.1.4.	1.1.5.

- 1.1.1. Which statement is incorrect regarding the heart?

- A. Its walls are made up of striated voluntary muscle
- B. It is situated in a space between the lungs called the mediastinum
- C. Internally it is divided into four chambers
- D. The upper atria are receiving chambers with relatively thin walls

- 1.1.2. The following events occur during a heartbeat.

- i. Semilunar valves forced open
- ii. Blood flows into atria
- iii. Both ventricles contract
- iv. Mitral and tricuspid valves open

Select the correct order in which these events occur:

- A. i, ii, iii, iv
- B. iii, iv, i, ii
- C. ii, iv, iii, i
- D. ii, iv, i, iii

- 1.1.3. Which of the following is not a characteristic of a capillary?

- A. Microscopic
- B. Narrow lumen
- C. Thin walls
- D. Large diameter

- 1.1.4. The fact that the left ventricular wall is thicker than the right shows that it:

- A. Pumps blood at a low pressure
- B. Pumps blood a greater distance
- C. Pumps blood through a smaller valve
- D. Has thickened over time

1.1.5. White blood cells responsible for engulfing foreign material are known as...

- A. Lymphocytes
- B. Phagocytes
- C. Antibodies
- D. Antitoxins

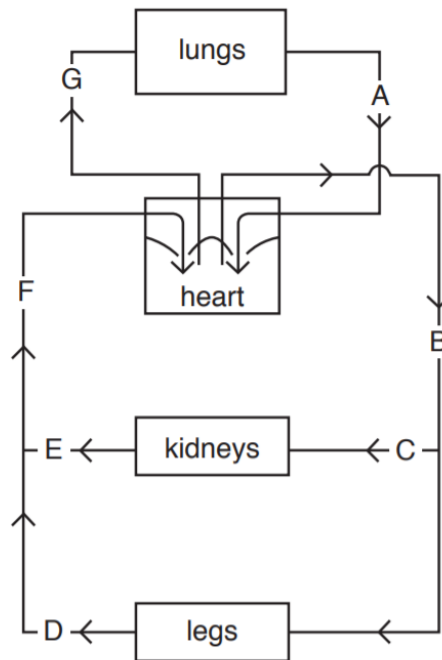
1.2 Indicate whether each of the descriptions in COLUMN I applies to A ONLY, B ONLY, BOTH A AND B OR NEITHER of the items in COLUMN II. Write **A, B, both** or **neither** in the column on the left. (5)

ANSWER	COLUMN I	COLUMN II
	1.2.1. Heart muscle supplied with oxygen and nutrients	A. Pulmonary artery B. Coronary artery
	1.2.2. Capillaries	A. Microscopic B. Easy diffusion of metabolites in and out of cells
	1.2.3. Oxygenated blood	A. All arteries B. Venules
	1.2.4. Systolic blood pressure	A. Atria are contracted B. Ventricles are relaxed
	1.2.5. Smooth muscle layer of blood vessels	A. Vasoconstriction B. Vasodilation

[10]

Question 2

2.1. Study the diagram below and answer the questions that follow.



2.1. By using the letters **describe** the blood flow of the pulmonary circuit. (4)

2.2. What vessel is represented by the letter C? (1)

2.3. What feature of the human circulatory system is being shown above? (2)

2.4. What structure will blood vessel D and F contain that the others do not? (1)

2.5. Provide a reason why the structure mentioned in 2.4. is needed. (2)

2.6. Name and differentiate between the two AV valves found in the heart.

(4)

[14]

Question 3

An investigation was performed to determine the effect of exercise on heart rate. A learner (A) recorded her resting heart rate three times and recorded it. She then recorded her heart rate after one minute of walking on the spot, one minute jogging on the spot and one minute of running as fast as possible. She then did the same for her friend (B). Their results are recorded below.

Activity	Heart rate (bpm) Learner A	Heart rate (bpm) Learner B
Rest	54	68
Rest	52	66
Rest	52	67
Walking	72	80
Jogging	94	112
Running	132	150

Immediately after the last activity they measured their pulse rate every minute for 5 minutes. They recorded this below.

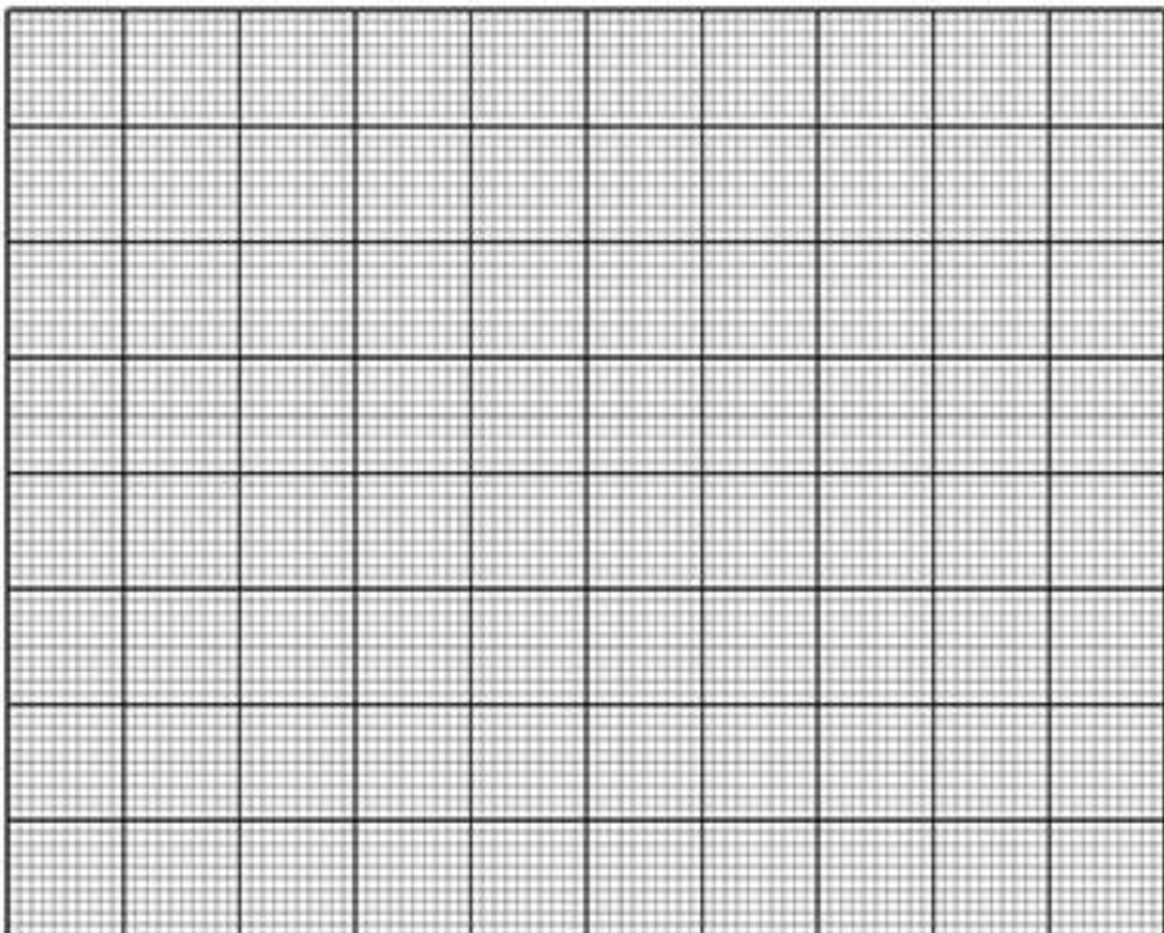
Time	Heart rate (bpm) Learner A	Heart rate (bpm) Learner B
1	110	137
2	95	120
3	72	102
4	54	84
5	54	72

3.1. With reference to the tables, provide an explanation for the increase in heart rate of both learners after jogging on the spot for one minute.

(2)

LIFE SCIENCES- TRANSPORT IN ANIMALS (2)

- 3.2. Which of the learners would you classify as the “fitter” of the two? Give a reason for your answer. (2)
-
-
- 3.3. Why do you think each learner recorded their resting heart rate three times? (2)
-
-
- 3.4. How long did it take for Learner A's heart rate to return to normal after running? (1)
-
- 3.5. State the dependent variable of this investigation. (2)
-
- 3.6. Draw an appropriate graph illustrating the results of **both** learners' heart rate for the five minutes after their last activity. (7)



[16]

TOTAL: 40 MARKS

[illegible]